TrES-1b

R-1004

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_161112` · created 2026-07-30T01:51:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457704.6443 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.022
Transit depth (Rp/Rs)^2	2.32 +/- 0.65 [%]
Orbital Inclination (inc)	89.94 +/- 0.82
Airmass coefficient 1 (a1)	1.433 +/- 0.027
Airmass coefficient 2 (a2)	-0.1185 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	None
Optimal Aperture	4.08
Optimal Annulus	8.92
Transit Duration (day)	0.1048 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.022 vs 0.1358 (deviation 0.61σ)
Duration fit vs published	0.1048 d vs 0.1042 d (deviation 0.22σ)
Mid-time O–C	4.1 min
Depth SNR	5.8
Residual scatter	1.72 %

Light curve

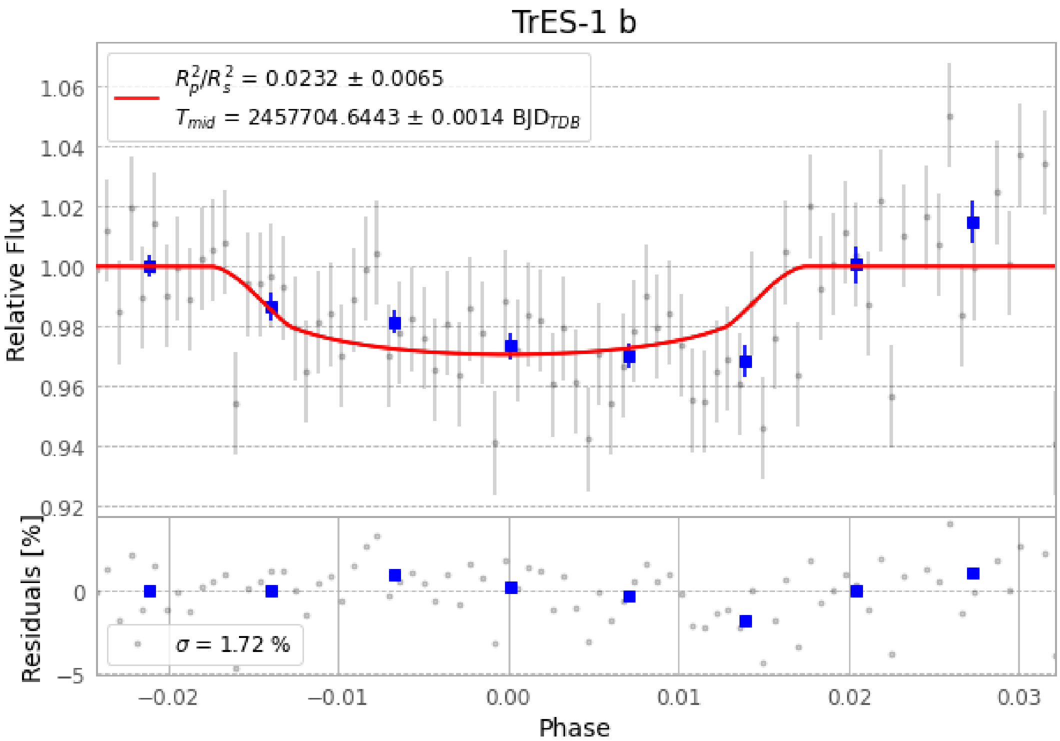

Figure 1 — FinalLightCurve_TrES-1 b_2016-11-11.png (SHA-256 d9c1942f494a01d1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 255], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 178c6645a2ecfd98…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

83 FITS frames (55 MB), `TRES-1161112014212.FITS` → `TRES-1161112054812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-161112.log (d29b14cd2191…), FinalLightCurve_TrES-1 b_2016-11-11.png (d9c1942f494a…), FinalLightCurve_TrES-1 b_2016-11-11.csv (d4d827fb810b…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)